

HUMAN ORGAN SYSTEMS

In the previous class you have studied about the cells, tissues, organs and functions of the major systems in the human body. You have also studied that different systems in the body co-ordinate their activities. Now you will study the structure and function of the digestive system.

Have you ever observed what happens inside our body, when we take a first bite of food? Any time we smell or taste or even think about good food, we feel some fluid inside the mouth. Where does this fluid comes from? What is the role of this fluid while eating?

In this Chapter you will learn about:

- Digestive System
- Disorder of Digestion (Constipation and Diarrhea)
- Respiratory System
- Common Diseases of Respiratory System

All the students will be able to:

- ✓ Describe various components of Human digestive system.
- ✓ Describe digestion and its importance.
- ✓ Describe how digestive system helps in the digestion of various kinds of foods.
- ✓ Identify common disorders of the digestive system.
- ✓ List the factors that lead to constipation and diarrhea, and the measures that can be taken to prevent them.
- ✓ Describe the mechanism of respiration in humans.
- ✓ Differentiate between breathing and burning processes.
- ✓ Identify the common diseases of respiratory system and discuss their causes and preventive measures.

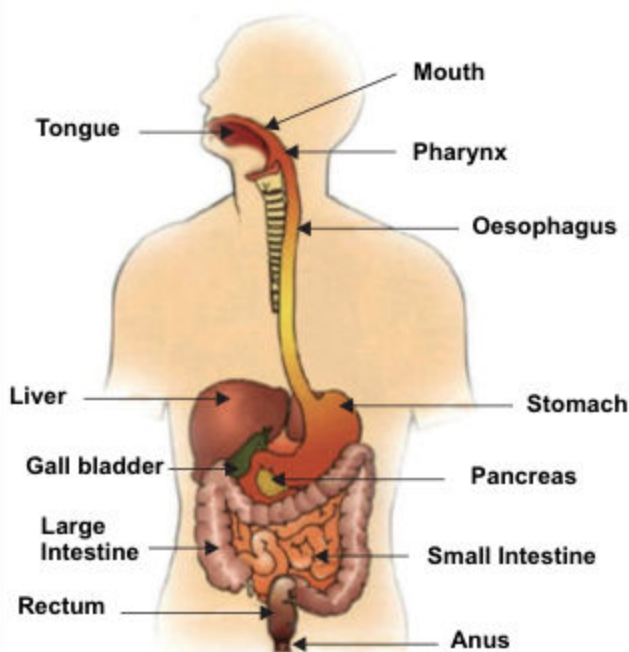


Figure 1.1 Human Digestive System

EXPLORING DIGESTION

Have you ever seen athletes take glucose solution before their races? Why?

HUMAN DIGESTIVE SYSTEM

- ✓ Describe various components of Human Digestive system.
- ✓ Describe digestion and its importance.
- ✓ Describe how digestive system helps in the digestion of various kinds of foods.

Take a bite from a piece of bread and chew it for some time.

You will find that the large pieces of bread breaks into smaller pieces. Is it necessary to convert the food into small pieces? Which part of our body performs this activity?

Food cannot go straight to the cells in the different parts of your body. It has to be broken down into smaller substances that can be absorbed into the body cells. Many organs of your body work together to achieve this goal. These organs work together to make up the digestive system and perform digestion.

“The large and complex food particles cannot enter into the cell without broken down by our body into smaller, simpler and diffusible form. This process of breaking down food into simple and diffusible molecules is called digestion.”

EXPLORING DIGESTION

Which organs of the digestive system come in contact with food, and what are some of their digestive functions?

DO YOU KNOW?

❖ Why do we need food?

Organisms need food because food contains:

- Useful chemical substances, called nutrients provides.
- Energy Important for activities for survival like movement, growth, maintains good health, etc.

❖ Digestion are of two types;

- **Physical digestion** is the mechanical breakdown of food into smaller pieces. This digestion provides large surface area for enzyme action.
- **Chemical digestion** is the break down food through enzymes (consists of protein molecules which speed up the rate of chemical reactions)

❖ Steps of digestion;

- There are five steps of digestion;
- Ingestion; taking of food inside the mouth.
- Digestion; breakdown of large food particles into small particles.
- Absorption; these small particles can easily diffuse through the walls of the alimentary canal into the blood stream.
- Utilization of food for cell processes.
- Egestion; elimination of undigested food.

INVESTIGATE

“YOU ARE WHAT YOU EAT”

Have you ever heard this expression? Explore the working of digestive system while eating junk food and healthy food.

Two groups of organs compose the digestive system: the alimentary canal and the accessory digestive organs.

The alimentary canal is a continuous tube, extending from the mouth to the anus. The length of this tract is about seven meters in a living person. It consists of mouth, pharynx, oesophagus, stomach, small-intestine, large intestine and anus.

The accessory digestive organs include the teeth, liver, gall bladder and pancreas.

Your food usually takes 24 to 30 hours to complete its journey inside your body. So when you are having your today's lunch, your digestive system is still working on the lunch that you ate the day before.

Let's take a trip round the alimentary canal.

Mouth:

The organ of digestive system in which both chemical and physical digestion takes place, in the mouth. The mouth consists of cheeks, lips, teeth, tongue, etc.

“Mechanical digestion in the mouth results from chewing, tearing and crushing by teeth, which helps food to get easily mixed with saliva. Saliva starts the chemical digestion of carbohydrates (starch). As a result, food is reduced to a soft and flexible semi-solid food called bolus, easily swallowed by tongue and move towards oesophagus.”

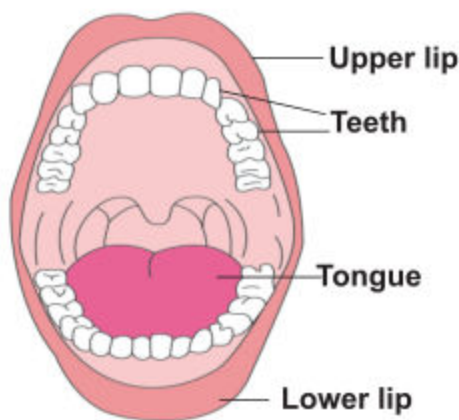


Figure 1.2 Mouth

Teacher Note: The teacher should show the model of human digestive system. Ask students questions related to main components of system during observation. Teacher also helps students to correctly identify the location of digestive organs and ask them to draw the diagram of the digestive system in their notebook.

Table 1.1: Summarizes the digestive activities in the mouth.

Structure	Activity	Result
Cheeks and lips	Keep food between teeth	Foods uniformly chewed
Walls of the mouth	Secrete saliva. Saliva contains enzyme.	Saliva softens, moistens and digests food Starch enzyme → Small glucose molecules
Tongue	Consists of taste buds	Identify different food tastes
Teeth	Cut, tear and crush the food	Solid foods reduced to smaller particles for swallowing

ACTIVITY 1.1 Investigating Oral Cavity:**What I need:**

- A piece of roti for each student.
- Precautionary measure:

Wash your hands thoroughly before starting the activity.



Figure 1.3 Traditional Roti

What to do:

S. No	Investigation of Oral Cavity	What I observed
1	How many teeth do you have?	
2	Why do you move your tongue during eating?	
3	What will happen if you do not move your tongue during chewing?	
4	Even before placing the piece of roti in your mouth your mouth became watery. Where this juice is coming from? Place your finger under your tongue and feel. Now place piece of roti in your mouth and hold it in mouth for few minutes.	

S. No	Investigation of Oral Cavity	What I observed
5	What happened to roti, when you hold it in your oral cavity? Why did it start to get softer? What did you taste? Did the taste changed? Did it become sweet? Did the roti get soft in your mouth even if you were not chewing? Yes/No, Why?	
6	Chew the roti and record your observations. Did you notice any change in taste after chewing for a while? Why?	
7	Why does a piece of roti become wet on chewing and it tastes sweet?	
8	Where do you think the digestion begins?	
9	What did you conclude from this activity about the beginning of chemical and mechanical digestion of food?	

Oesophagus (food pipe):

Oesophagus acts as a pathway and transports the food from pharynx to stomach. Bolus is pushed from oesophagus by a wave like pattern movement. No digestion takes place here.

Stomach:

From the oesophagus bolus enters into stomach. Stomach is a J-shaped muscular bag. The walls of the stomach release gastric juice. This juice contains

DO YOU KNOW?

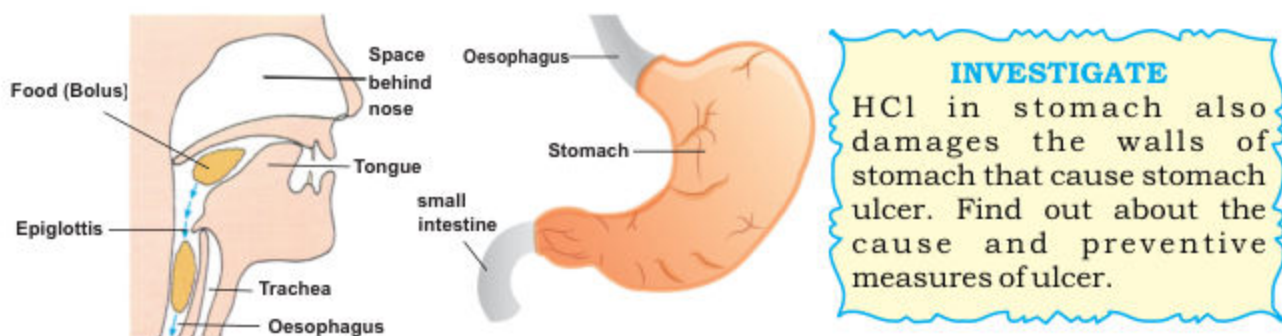
❖ Pharynx and Swallowing:

Pharynx is a funnel shaped muscular organ that connects mouth to the oesophagus (digestive organ) as well as larynx (respiratory organ). Muscular contraction of the pharynx helps to propel food into the oesophagus and then into the stomach. This movement of food is achieved by the act of swallowing.

❖ What does epiglottis perform during swallowing?

As larynx is in front of the oesophagus so there is a possibility food enters into the respiratory track during swallowing. This is usually prevented by a flap like structure, epiglottis.

an acid called hydrochloric acid (HCl), water and enzyme. The enzyme converts the large protein molecules into small protein molecules. HCl softens the food and also kills the germs which are present in the food.



INVESTIGATE

HCl in stomach also damages the walls of stomach that cause stomach ulcer. Find out about the cause and preventive measures of ulcer.

Figure 1.4 Food swallowing process and Stomach

Mechanical digestion in the stomach results from mixing waves, in which bolus is completely mixed with gastric juice.

Gastric juice starts the chemical digestion of protein. As a result, bolus is converted into a semi liquid food called chyme.

Food is held in the stomach for several hours. Gradually the stomach empties its contents into the small intestine.

Table 1.2: Summarizes the digestive activities in the stomach.

Structure	Activity
Muscular walls	Contraction of muscles generates mixing waves
HCl	Softens the food Kills the germs present in the food
Enzyme	Large protein molecules enzyme → Small protein molecules

DO YOU KNOW?

❖ Liver and pancreas:

Both the liver and pancreas play a role in digestion. Without their secretions digestion remains incomplete. As liver contains gall bladder which stores bile. However the food does not pass through these structures that's why they are not the part of alimentary canal.

❖ Obesity:

Obesity is the most common nutritional disorder. If energy input as a result of eating is greater than energy used, the extra energy is stored as fat so become overweight or obese. An obese person is much more likely to suffer from high blood pressure, cancer, stomach disorder.

EXPLORING DIGESTION

If food moves down the oesophagus because of gravity or by wave like movement then what would happen if a person stand on his head?

Small Intestine:

Small intestine is a long, highly coiled narrow tube of six meter average length. As the chyme enters in the small intestine, it receives bile from the liver and pancreatic juice from the pancreas.

Bile is a greenish-brown juice, secreted by liver and stored in gall bladder. It helps to breakdown large sized fats into smaller pieces so that fats can be digested faster by enzymes.

Pancreatic juice is secreted by pancreas. This juice contains enzymes which help to digest proteins, fats and carbohydrates.

The small protein molecules are now broken down into amino-acids; simpler and diffusible form of proteins. The semi-digested carbohydrates are converted into diffusible sugars and fats are broken down into glycerol and fatty acids. The muscular action of small intestine mixes the food with bile and pancreatic juice.

Here the digestion is completed. At the end of the small intestine, the digested, soluble and diffusible components of food now diffuse through the walls of the small intestine into the blood stream. This process is called absorption.

Absorption only occurs in small intestine as its wall consists of many folds that provides larger surface area. Its walls are completely surrounded by capillaries for maximum absorption.

The absorbed food is utilized in the body, like glucose used in the respiration process. Amino-acids, fatty acids and glycerol are used in the formation and growth of cells.

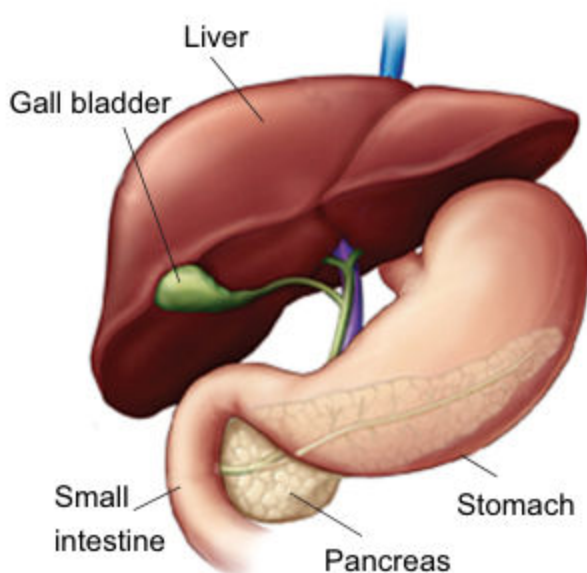


Figure 1.5 Small intestine along with liver and pancreas

Table 1.3: Summarizes the digestive activities in the small intestine.

Structure	Activity	Result
Small intestine	Muscular activity	Mixes the food completely with juices. Also speed up the digestion
Liver	Bile	Breakdown fats into smaller pieces so that fats can be digested faster by enzymes
Pancreas	Pancreatic juice	Small protein molecules <u>enzyme</u> → Amino acids Semi-digested carbohydrates <u>enzyme</u> → Diffusible sugars Small fats pieces <u>enzyme</u> → Fatty acids and glycerol

Large Intestine:

The undigested, non-diffusible and insoluble components of food moves towards the last part of digestive system called large intestine.

It is shorter but much broader than the small intestine. It is a tubular structure, which is on average 1.5m long with a large diameter; here the water and mineral salts are absorbed.

The minerals are absorbed with the help of some friendly bacteria which live here. All the non-diffusible components of food move towards **rectum**. Here the undigested food is converted into faeces which are removed from the body by an opening called **anus**.

DO YOU KNOW?

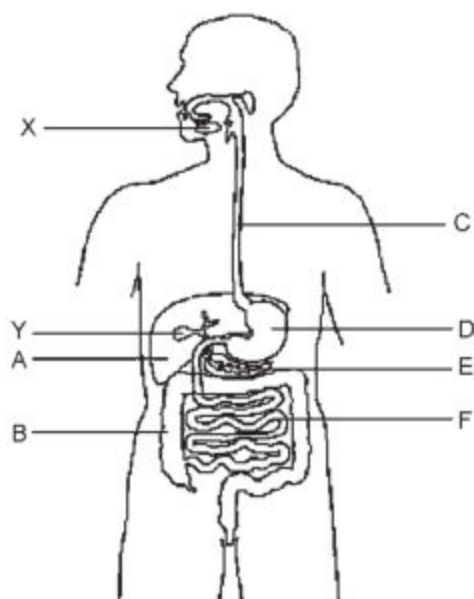
❖ Appendicitis:

Large intestine contains finger like projection called appendix. It plays no role in digestion but can become infected and inflamed that cause appendicitis.

ACTIVITY 1.2:

Imagine you are a medical student. Your teacher has assigned you to analyse the human digestive system by using graphic organiser. Here are the things you may need to do.

- Carefully observe the model of human digestive system.
- Label the diagram. Also draw a neat diagram to learn how these organs link to each other to form a system.
- Identify each organ of the system.
- Write down the function of each of the organs.
- Discuss with your teacher what would happen if the organs were missing.



Space for Labelling :

A =

B =

C =

D =

E =

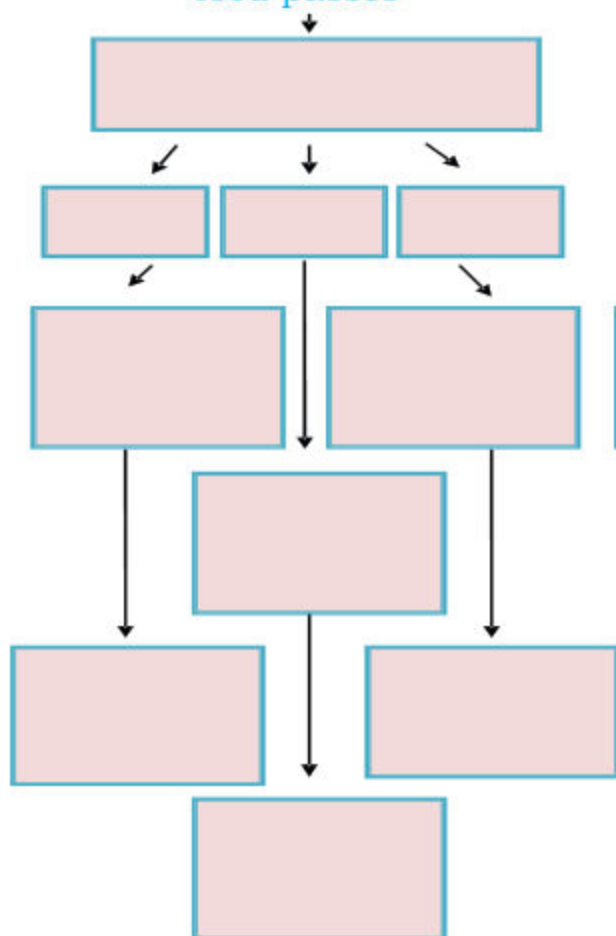
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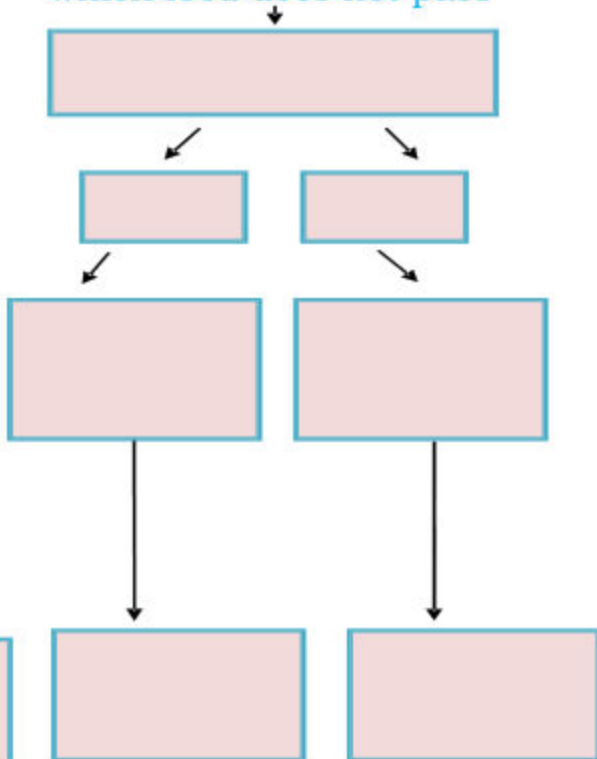
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Human digestive system

A tube through which
food passes



Structures through
which food does not pass



DISORDERS OF DIGESTION

- ✓ Identify common disorders of the digestive system.
- ✓ List the factors that lead to constipation and diarrhea, and the measures that can be taken to prevent them.

INVESTIGATE

Find out the causes of diarrhea other than contaminated food.

What happens when we eat any contaminated food? Does it affect our digestive system? It may cause any damage to our system due to which digestive system does not function properly. Some of the digestive tract disorders are diarrhea, food poisoning, vomiting, ulcer, constipation, obesity and malnutrition. Most common disorders are discussed below.

Diarrhea:

Have you ever heard about loose motion? It is medically termed as diarrhea. It occurs due to fast downwards rhythmic action of digestive tract. The reasons include presence of micro-organisms in the contaminated food and water.

It might also occur because of abnormal secretion of juices. This may cause loss of body water (dehydration) and salts. This loss disturbs the normal body functions, consequently the person feels weak.

Constipation:

It is the reverse process of diarrhea in which the rhythmic action of digestive tract becomes slow. The undigested food remains in rectum for long period where extra absorption of water occurs. As a result the faeces turn into hard mass which are difficult to pass out and cause pain.

DO YOU KNOW?

How diarrhea and constipation can be avoided?

❖ Diarrhea

- Avoid drinking contaminated water.
- Always use boiled/filtered water.
- Use properly washed fruits and vegetables.
- Wash your hands before eating food.
- Use properly washed utensils.
- Don't use prolonged improperly stored food.

❖ Constipation

- Drink large amount of water.
- Use food containing dietary fibre.
- Process of egestion must be done properly on daily basis.

ACTIVITY 1.3: ROLE PLAY ACTIVITY

- Teacher will form groups of six students in a class.
- Each disorder (either constipation or diarrhea) should be taken as a character.
- Ask each member to find out the (causes, prevention and control) among the group members.
- Use headband or tags (made from chart paper) to show the name of the role you are playing.
- Learn your dialogues.
- Each group will present their role play in front of whole class.

RESPIRATORY SYSTEM

- ✓ Describe the mechanism of respiration in humans.
- ✓ Differentiate between breathing and burning processes.

EXPLORING DIGESTION

Why does respiration take place in every living cell all the time?

Do you think, while you are sleeping, you probably don't need energy?

When we consider ourselves, we only link the word energy with doing exercise, playing games, etc. Your body is doing a lot of work while sleeping, like breathing, beating of heart, movement of food in alimentary canal, etc. All these processes require energy.

In the previous class you have studied how living organisms produce energy. You have also studied the major organs of the respiratory system that provides oxygen in the human body. Now you will study the mechanism of respiration in humans.

What is respiration? How respiration occurs?

Let's discuss the whole process.

“Respiration is a biochemical process during which simple carbohydrates like glucose are broken down to release energy.”

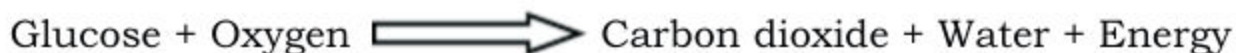
Every living organism needs oxygen and food for respiration. Food is converted into glucose that comes from the digestive system. We obtain oxygen from

INVESTIGATE

Respiration is an involuntary process. Justify this statement with scientific reasons.

our respiratory system. Then our circulatory system transport oxygen and food to the cell, where glucose combines with oxygen to release energy. In this process carbon dioxide and water are given off as waste products.

The whole process of respiration can be represented by word equation:



Nostrils
↓
Pharynx
↓
Larynx
↓
Trachea
↓
Bronchi
↓
Bronchioles
↓
Alveoli

Oxygen diffuses from alveoli to blood. From blood oxygen then transfer to all body cell.

DO YOU KNOW?

How oxygen enters inside your body?

We get oxygen through respiratory system. The flow chart below shows the pathway of oxygen within the respiratory system:

Respiratory System

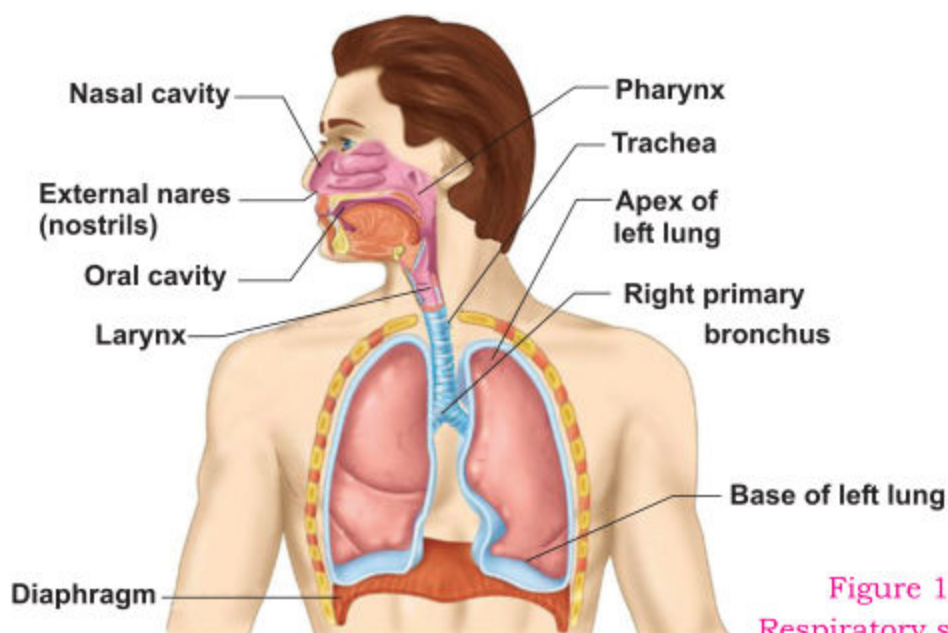


Figure 1.6
Respiratory system.

Respiration involves two processes

- Breathing
- Gaseous exchange

EXPLORING RESPIRATION

How do divers breathe under water as they have lungs not gills?

What is breathing?

Can you stay alive without food or water for about one day? The expected answer is “yes we can”. Can you stay alive without breathing even for a short period of time? Your answer should be “no we cannot”.

We breathe through our respiratory system. As you observed in figure 1.7, the main organs of respiratory system through which oxygen enters the body. In order to make breathing possible the function of following parts is very important.

- Ribcage, collection of bones that protect the delicate lungs.
- Diaphragm, a muscular sheet.
- Thoracic cavity or chest cavity, protects respiratory system.

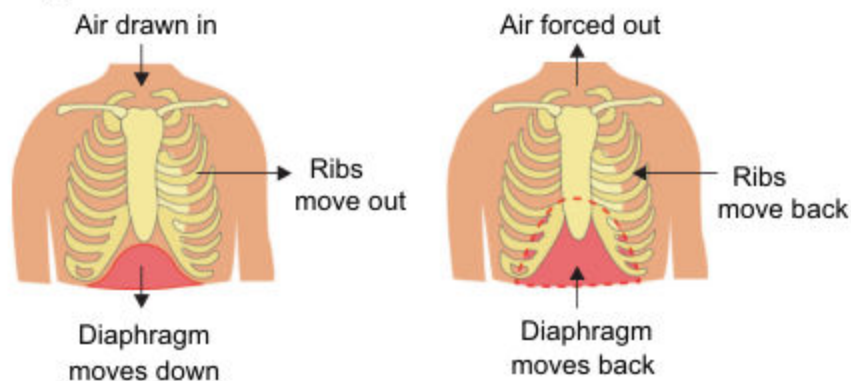
Breathing is a physical process in which oxygen is taken inside the body and carbon dioxide is given out from the body.

INVESTIGATE

Find out the process of breathing while coughing and sneezing.

Breathing requires two processes

- The taking in of oxygen is called inhalation.
- The giving out of carbon dioxide is called exhalation.



When we breathe in (inhalation), diaphragm flattens, cause the ribcage to move out thus enlarges chest cavity.

When we breathe out (exhalation), diaphragm curves upwards, cause the ribcage to move back thus decreasing the size of chest cavity.

Figure 1.7 Illustrates the process of breathing.

What is gaseous exchange?

With the action of diaphragm, ribcage and thoracic cavity, oxygen from atmosphere comes inside the alveoli (lungs). How does oxygen from the alveoli diffuse in the blood (capillaries)? How does carbon dioxide from the blood diffuse inside the alveoli?

“The exchange of gases between alveoli and blood capillaries is called gaseous exchange.”

You have already studied the structure of alveoli.

Alveoli have a microscopic structure consisting of single cellular layer.

They are completely surrounded by blood capillaries. When oxygen enters inside the alveoli, the diffusion process occurs. The oxygen from alveoli (high level) diffuses in blood capillaries (low level). Whereas carbon dioxide from the blood capillaries (high level) diffuses in alveoli (low level).

When carbon dioxide reaches the alveoli, it moves towards bronchioles, bronchi, trachea, larynx, pharynx and finally leaves the body through nostrils. This is called exhalation as we discussed earlier. Blood transports oxygen to the cells where it reacts with food to give energy.

INVESTIGATE

Record your breathing rate while you are watching TV and after doing two minutes exercise.

DO YOU KNOW?

Is it possible to respire without oxygen?

Small microorganisms like fungi and bacteria respire without oxygen. This process is called anaerobic respiration. If oxygen is used in the respiration process it is called aerobic respiration like humans.

Anaerobic respiration is used in the production of vinegar, cheese, bread, etc.

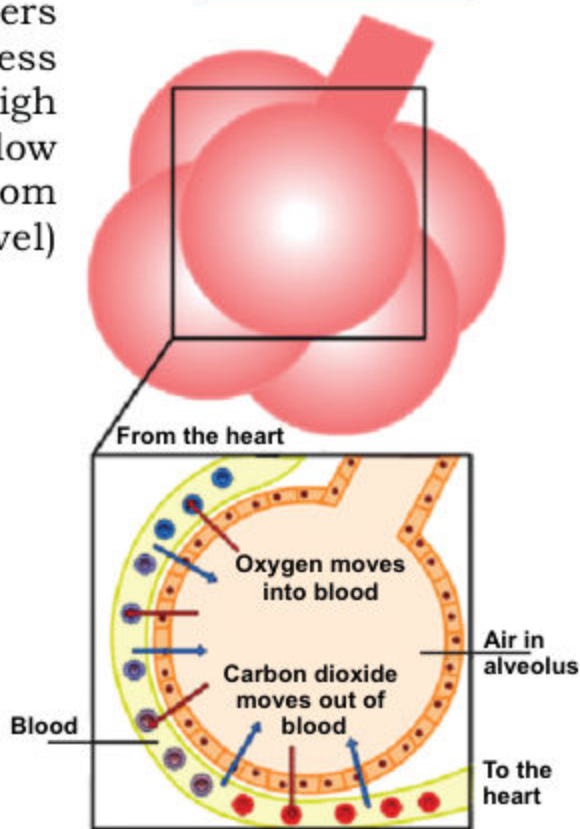


Figure 1.8 Gaseous exchange between alveoli and blood capillaries

ACTIVITY 1.4 To demonstrate the process of breathing.

What I need:

- Plastic bottle.
- Cutter.
- 3 balloons.
- Cork.
- 2 Straws.

What to do:

1. Take a plastic bottle, cut into half and use the upper-half of the bottle.
2. Put a rubber sheet (made by a balloon) at wide open part of bottle and seal it.
3. Put a cork at the mouth of bottle.
4. Make a hole to insert straw in it.
5. Fit a balloon at one end of straw which is inside the bottle.
 - a) What happens when the rubber sheet is pushed down and pushed up?
 - b) Note your observations and explain the biological process which has been shown in this model.

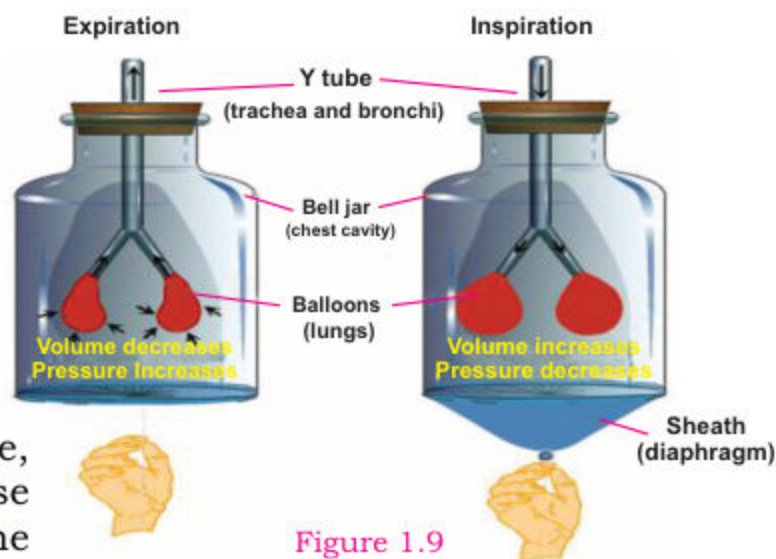


Figure 1.9
Experiment shows breathing process

What I observed:

INVESTIGATE

Find out different ways of breathing process and gaseous exchange in animals?

What I concluded:

ACTIVITY 1.5 Lime water test:

To find out whether carbon dioxide is given off during respiration.

What I need:

- Test tube.
- Lime water.
- Straw.

What to do:

1. Take a test tube containing lime water.
2. With the help of the straw forcefully blow the expired air, inside the test tube.
3. Put a cork at the mouth of test tube.
4. Shake the test tube containing your expired air.

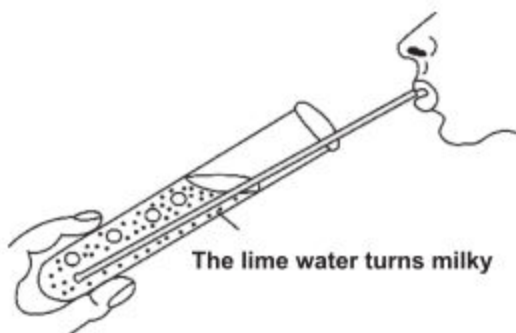


Figure 1.10 Carbon dioxide is exhaled during respiration

What I observed:**Activity questions:**

1. Does the colour of the lime water change?
2. Why the lime water only gives that colour?

What I concluded:

Teacher Note: The teacher needs to arrange the apparatus for experiments. Also allow the students to perform the experiments in the presence of the teacher in the class.

Differences between Breathing and Burning Process

Do breathing (respiration) and burning differ from each other? Do you know breathing provides oxygen to the burning process to burn the food? Let's discuss more differences between them in the class.

Table 1.4 summaries the differences between breathing and burning process.

Breathing	Burning Process
It is a physical process as it forces air into and out of the lungs.	It is a chemical process as it burns food in the presences of oxygen.
It occurs in the respiratory system.	It takes place in all living cells.
It requires energy.	It releases energy.
Enzymes are not involved.	Various enzymes are required.

RESPIRATORY DISORDERS

- ✓ Identify the common diseases of respiratory system and discuss their causes and preventive measures.

Some common disorders of respiratory tract are cough, common cold, asthma, tuberculosis, pneumonia, and lung cancer. Here we discuss common cold, tuberculosis and asthma.

Cold:

The cold is a common disorder of the upper respiratory tract (nose and throat). The main cause of the cold is environmental pollution.

DO YOU KNOW?

Preventive measures of respiratory disorders

We can prevent ourselves from these disorders by eating fresh fruits and vegetables. Do regular exercise. Properly clean yourselves. Live in a pollution free environment. Do proper vaccination. Do not smoke.

sore throat, cough, congestion, mild headache, sneezing, watery eyes, low grade fever, mild fatigue, etc. There is no cure for the cold. It can be relieved by symptomatic treatment such as taking over the counter medications (throat lozenges, throat sprays, cough drops and cough syrups). Gargling with salt water may help those with a sore throat.

Tuberculosis (T.B):

Tuberculosis is a disorder of lower respiratory system (lungs). It is an infectious bacterial disease caused by *Mycobacterium tuberculosis*, in which the lung is damaged resulting in cough and fever. It is transmitted from person to person via nasal secretion and sputum. The symptoms of the active T.B of the lungs are coughing, sometimes sputum with blood, chest pains, weakness, weight loss, fever and night sweats. T.B is treatable with proper medical attention. It is more common in poor people. Malnutrition and poor living conditions facilitate *Mycobacterium* to grow.

Asthma:

Asthma is a serious chronic disease of the respiratory airways. In asthma bronchial tubes are always inflamed (swelling). It is an inherited disease, which means it transfers from parents to their offspring. It is an allergic response against cold, humidity, pollution, etc. Patients usually suffer from shortness of breath. Other symptoms are chest tightness or pain, chronic coughing and wheezing (whistling sound). It is not easy to cure asthma, but symptoms can be controlled with effective medication. Use inhaler and keep it with you all the time. It is better for the asthma patients to avoid congested places.

Summary

Mouth starts carbohydrates digestion. Also break food into pieces by teeth.

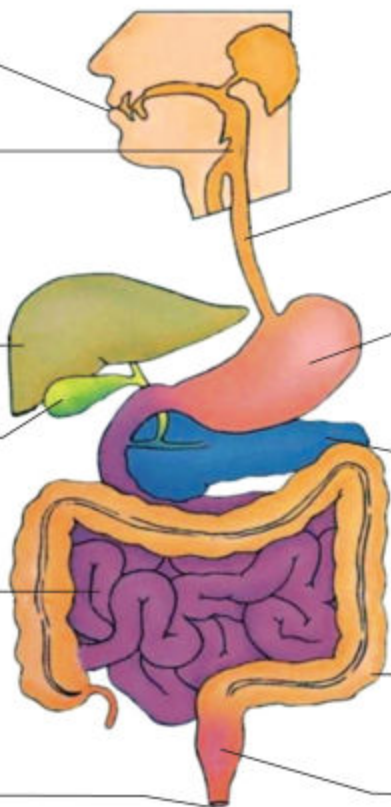
Pharynx swallows food.

Liver produces bile to break fat molecules into small pieces.

Gall bladder stores bile

Small intestine completes the digestion of carbohydrates, fats and protein. It helps in absorption of food.

Anus egests faeces.



Oesophagus transports food

Stomach starts protein digestion. HCl kills germs.

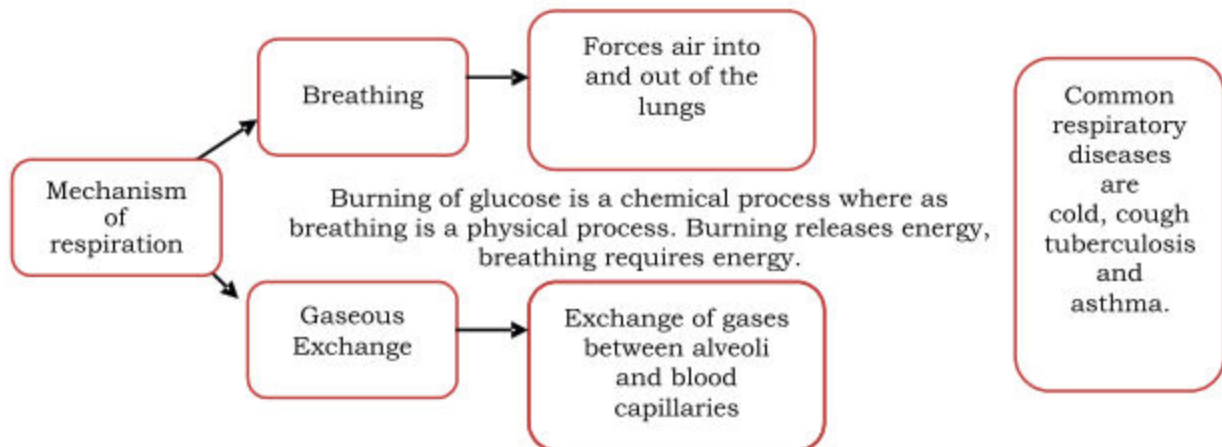
Pancreas digests carbohydrates, fats and protein.

Large intestine absorbs water and minerals.

Rectum stores faeces.

If rhythmic action of digestive tract becomes slow cause constipation whereas if rhythmic action of digestive tract becomes fast cause diarrhea.

HUMAN RESPIRATORY SYSTEM



Review Exercises

1. Circle the best answer.

- i) Which is secreted by the liver and stored in the gallbladder?
(a) Saliva (b) Gastric juice
(c) Bile (d) Pancreatic juice
- ii) Where chemical digestion of proteins start?
(a) Mouth (b) Oesophagus
(c) Stomach (d) Small intestine
- iii) Which line shows the correct order of alimentary canal?
(a) Mouth → Oesophagus → Small Intestine → Stomach
(b) Pharynx → Stomach → Pancreas → Liver
(c) Pharynx → Liver → Pancreas → Small Intestine
(d) Mouth → Oesophagus → Stomach → Small Intestine
- iv) Which structure prevents the food from entering the larynx during swallowing?
(a) Pharynx (b) Oesophagus
(c) Epiglottis (d) Tongue
- v) Which option correctly shows the difference between burning and breathing process?

	Burning Process	Breathing Process
(a)	It is a physical process as it forces air into and out of the lungs	It is a chemical process as it burns food in the presence of oxygen
(b)	It takes place in all living cells	It occurs in the respiratory system
(c)	It requires energy	It releases energy
(d)	Enzymes are not involved	Various enzymes are required

2. Explain why:

- i) Liver and pancreas are not considered as the parts of alimentary canal.
- ii) Fats are not digested in mouth and stomach.
- iii) Absorption only occurs in small intestine.
- iv) Breathing is a physical process.
- v) Pollution causes respiratory disorders.
- vi) Don't speak while eating food.

3. Give brief answer to the following questions:

- i) What is mechanical digestion? Discuss the processes of mechanical digestion that takes place along the alimentary canal.
- ii) Describe the process of chemical digestion of a piece of potato that has been fried in oil, as it passes through following organs. Support your answer with enzyme word equations.
 - (a) Mouth
 - (b) Stomach
 - (c) Small intestine
- iii) What preventive measures should be taken to avoid respiratory disorders?
- iv) Discuss the adaptation of alveoli for gaseous exchange process.
- v) Draw a flow chart to show the digestion of food in the human alimentary canal.

- 4. (a) Draw a labelled diagram of alimentary canal. Also draw the correct position of pancreas and liver in the digestive system.
- (b) Draw a labelled diagram of gaseous exchange process in human body.

PROJECT

Object:

To investigate the process of respiration in an animal (Cockroach).

What I need:

- i) Two conical flasks tightly covered with rubber bung.
- ii) One small animal like cockroach or snail.
- iii) Formalin chemical.

What to do:

- i) Take two conical flasks. Place a live cockroach or snail in one flask and a dead cockroach or snail, that has been soaked in formalin to prevent decay, in the other flask.
- ii) Close the mouth of each flask with rubber bung and make sure that the apparatus is airtight.
- iii) Leave the flasks for three hour.
- iv) After three hour, introduce a small lighted candle into each flask.
- v) After introducing the candle, close the mouth of the flask tightly.
- vi) Note the time taken for the candle flame to go off.

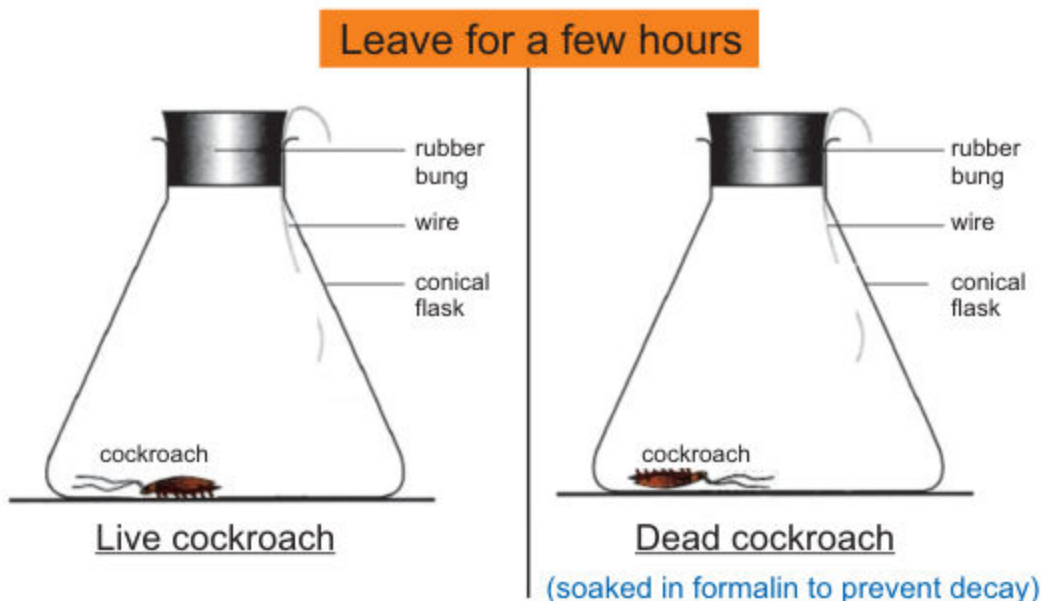


Figure Experiment demonstrates respiration in animal

What I observed:

Activity questions:

- i) Give an explanation for your observation.
- ii) Why we use dead animal in this experiment?
- iii) Why we use airtight conical flasks?

What I concluded: